Chapter 4: Electrostatics

The following subjects will be discussed

- Gauss's Law and its Applications.
- The Electric potential.

Gauss's Law

Gauss law relates the electric field to the charge producing it. Knowing the field, we can calculate the charge.

$$Q_{enclosed} = \psi_{net}$$

Total charge enclosed inside a 3D closed surface.

Net Flux. Number of field lines crossing a closed surface.

$$\psi_{net} = \oint \vec{D} \cdot \overrightarrow{dS}$$

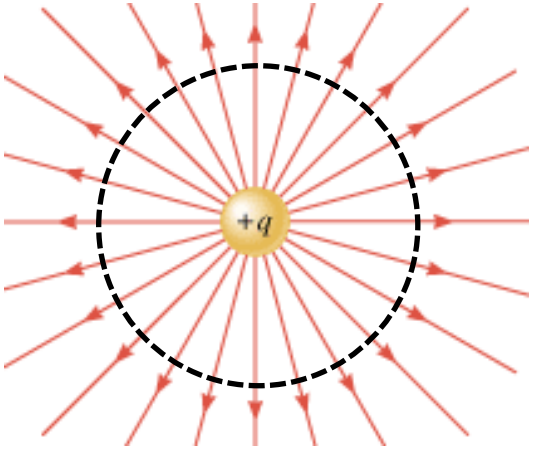
Differential area

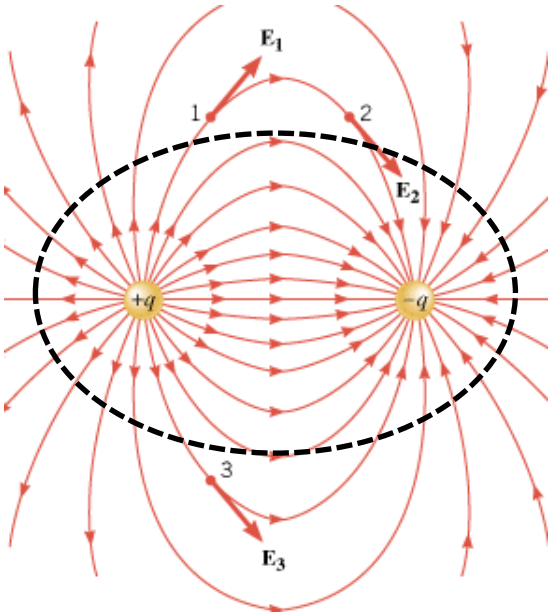
Closed surface integral

Electric flux density generated by charge

Dot product

$$\vec{D} = \varepsilon \vec{E}$$





Using divergence theorem

$$\oint_S \vec{D} \cdot \overrightarrow{dS} = \int_V \vec{\nabla} \cdot \vec{D}\ dv = Q \quad \text{and} \quad Q = \int_V \rho_v\ dv$$

The following equation is obtained

$$\vec{\nabla} \cdot \vec{D} = \rho_v$$

Flux Calculations

Example. Calculate the net electric flux out of the cube shown below given the following electric field in free space. $\vec{E} = 4\,\vec{a}_x - 3(y^2 + z)\,\vec{a}_y$ (V/m)

$$\varphi_{net} = \varepsilon_o \oint \vec{E} \cdot \overrightarrow{dS} = \varphi_{top} + \varphi_{bottom} + \varphi_{left} + \varphi_{right} + \varphi_{front} + \varphi_{back}$$

$$\varphi_{right} = \varepsilon_o \oint \vec{E} \cdot \overrightarrow{dS} = \varepsilon_o \int_0^2 \int_0^2 -3(y^2 + z)\,dx\,dz \quad (y = 2) = \varepsilon_o \int_0^2 \int_0^2 -3(4 + z)\,dx\,dz$$

$$\varphi_{right} = \varepsilon_o \int_0^2 \int_0^2 (-12)\,dx\,dz + \varepsilon_o \int_0^2 \int_0^2 (-3z)\,dx\,dz$$

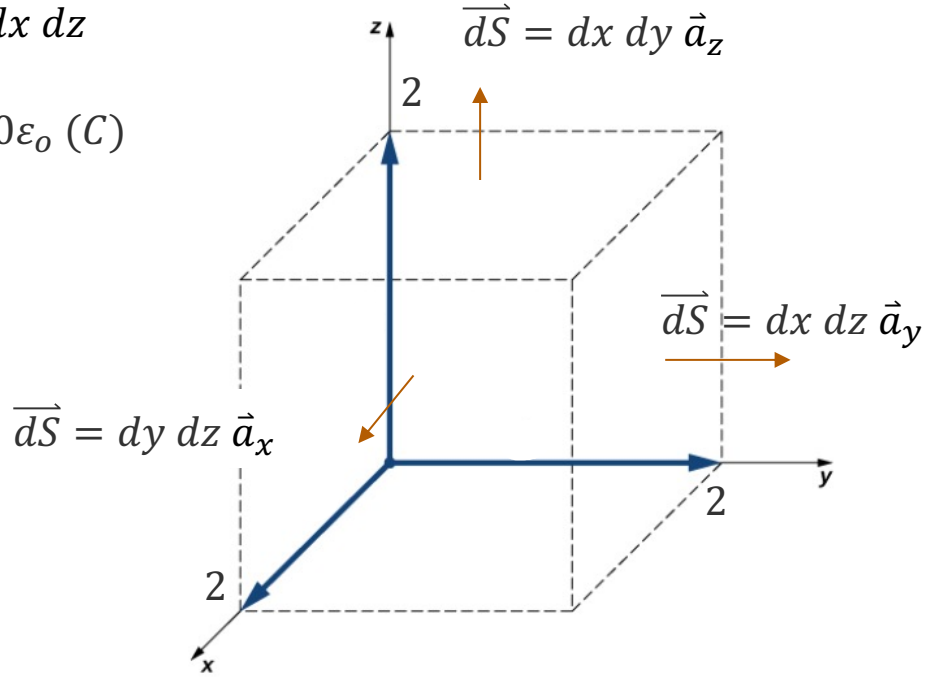
$$\varphi_{right} = \varepsilon_o(-12)(2)(2) + \varepsilon_o(-3)(2)\left(\frac{4}{2}\right) = -60\varepsilon_o\ (C)$$

$$\varphi_{left} = \varepsilon_o \int_0^2 \int_0^2 -3(y^2 + z)\,dx\,dz = 12\varepsilon_o\ (C) \quad (y = 0)$$

$$\varphi_{front} = \varepsilon_o \int_0^2 \int_0^2 4\,dy\,dz = 16\varepsilon_o\ (C)$$

$$\varphi_{back} = \varepsilon_o \int_0^2 \int_0^2 -4\,dy\,dz = -16\varepsilon_o\ (C)$$

$$\varphi_{net} = -48\varepsilon_o\ (C)$$



Flux Calculations

Example. Calculate the total charge enclosed in a cylinder with axis coincident with the z-axis, between z=0 and z=5 m, and radius of 3 m. The electric flux density is given by $\vec{D} = 2\rho z\cos^2\varphi\ \vec{a}_\rho - \rho z\sin\varphi\cos\varphi\ \vec{a}_\varphi + \rho^2\cos^2\varphi\ \vec{a}_z$ (C/m^2)

$$Q_{enclosed} = \varphi_{net} = \oint \vec{D}\cdot\overrightarrow{dS}$$

$$\varphi_{around} = \int_0^{2\pi}\int_0^5 (2\rho z\cos^2\varphi)(\rho\, d\varphi\, dz)$$

$$= 2\rho^2\int_0^5 z\, dz\int_0^{2\pi}\int_0^5 \cos^2\varphi\, d\varphi \qquad (\rho = 3)$$

$$= 2(3^2)\left(\frac{5^2}{2}\right)\left[\frac{\varphi}{2} + \frac{\sin 2\varphi}{4}\right]\Bigg|_0^{2\pi} = 225\,\pi \qquad (C)$$

$$\varphi_{top} + \varphi_{bottom} = 0$$

$$Q_{enclosed} = 225\,\pi \quad (C)$$

Field Calculations using Gauss Law

Gauss law can be applied to find the field due to a charged structure. However, this is possible only under the following conditions.

Application of Gauss law for field calculation. Enclose the charged structure with an imaginary suitable Gaussian surface.

$$Q_{enclosed} = \oint \vec{D} \cdot \overrightarrow{dS} = \varepsilon \oint \vec{E} \cdot \overrightarrow{dS}$$

(1) Closed
(2) Field is either completely normal or completely tangential
(3) Field is constant all over surface.

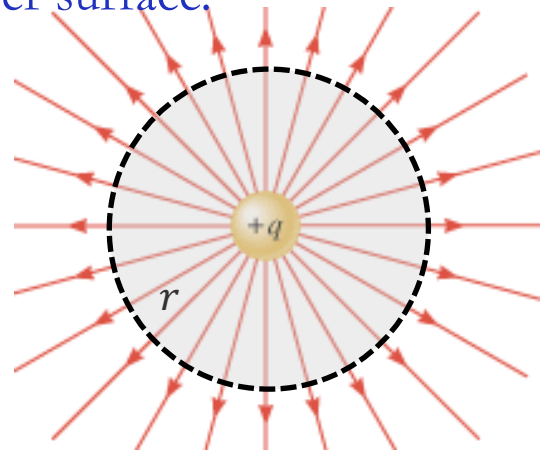
Example. A point charge

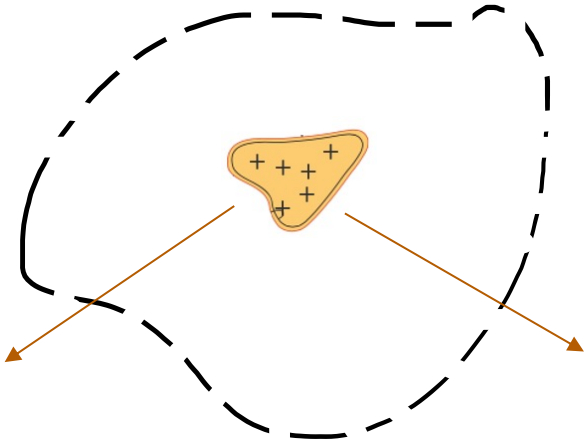
$$Q_{enclosed} = q$$

$$\oint \vec{D} \cdot \overrightarrow{dS} = D \oint dS$$

$$\oint \vec{D} \cdot \overrightarrow{dS} = D\,(4\pi r^2)$$

$$q = D\,(4\pi r^2) \Rightarrow \vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{q}{r^2}\,\vec{a}_r$$





Can be applied to structures that possess symmetry.

Field Calculations using Gauss Law

Example. An infinite line with uniform charge density ρ_L (C/m).

$$Q_{enclosed} = \varphi_{net} = \oint \vec{D} \cdot \overrightarrow{dS}$$

$$Q_{enclosed} = \rho_L L$$

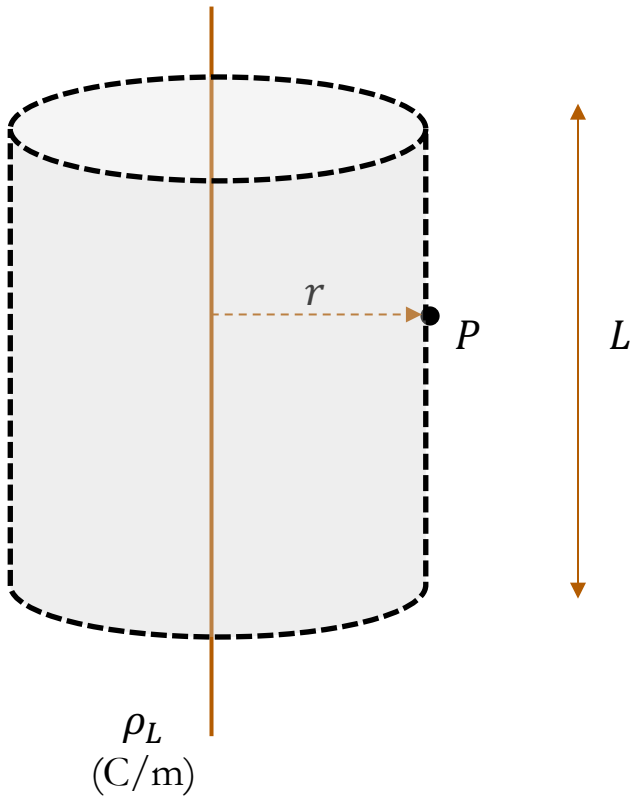
$$\varphi_{net} = \varphi_{top} + \varphi_{bottom} + \varphi_{side}$$

$$\varphi_{side} = \oint \vec{D} \cdot \overrightarrow{dS} = D\,(2\pi r L)$$

$$\rho_L\, L = D\,(2\pi r L)$$

$$D = \frac{\rho_L}{2\pi\, r}$$

$$\vec{E}_p = \frac{\rho_L}{2\pi\varepsilon\, r}\vec{a}_r$$



Field Calculations using Gauss Law

Example. An infinite flat surface with uniform charge density σ (C/m^2).

$$Q_{enclosed} = \varphi_{net} = \oint \vec{D} \cdot \overrightarrow{dS}$$

$$Q_{enclosed} = \rho_s A$$

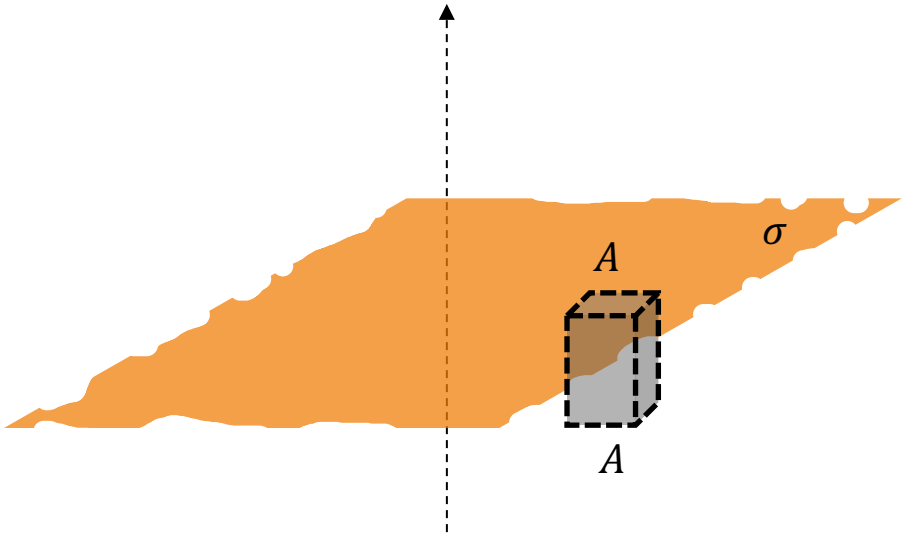
$$\varphi_{net} = \varphi_{top} + \varphi_{bottom} + \varphi_{sides}$$

$$\varphi_{top} = \varphi_{bottom} = \oint \vec{D} \cdot \overrightarrow{dS} = D\,(A)$$

$$\rho_s A = 2D\,(A)$$

$$D = \frac{\rho_s}{2}$$

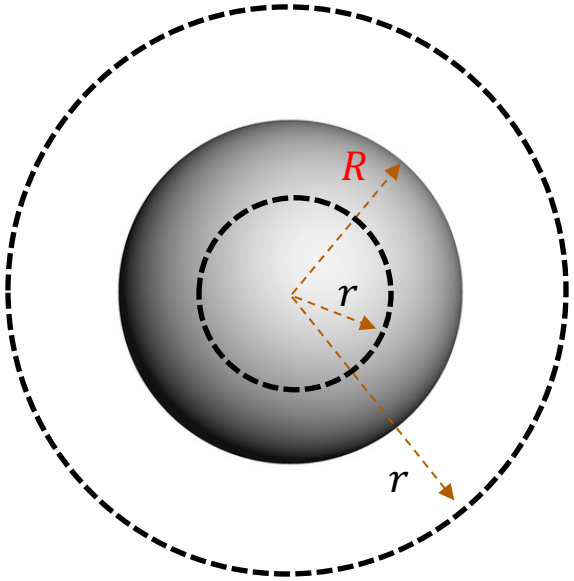
$$\vec{E}_p = \frac{\rho_s}{2\varepsilon}\vec{a}_n$$



Field Calculations using Gauss Law

Example. A solid non-conducting sphere of radius R with uniform volume charge density ρ_v (C/m^3) in free space.

$$Q_{enclosed} = \varphi_{net} = \oint \vec{D} \cdot \overrightarrow{dS}$$



1) For $r \leq R$ (points inside the sphere)

$$Q_{enclosed} = \rho_v \frac{4\pi}{3} r^3$$

$$\varphi_{tot} = \oint \vec{D} \cdot \overrightarrow{dS} = D\,(4\pi r^2)$$

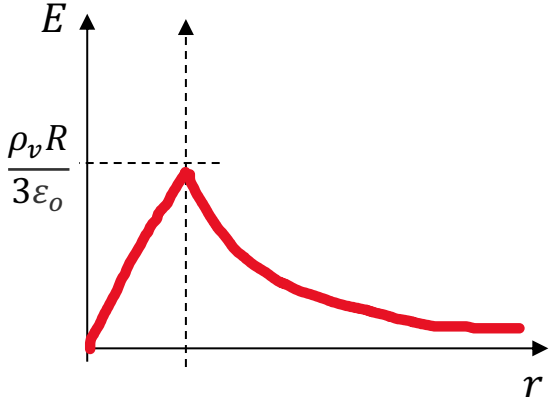
$$\rho_v \frac{4\pi}{3} r^3 = D\,(4\pi r^2) \qquad \vec{E} = \frac{\rho_v}{3\varepsilon_o} r\,\vec{a}_r$$

2) For $r \geq R$ (points outside the sphere)

$$Q_{enclosed} = \rho_v \frac{4\pi}{3} R^3$$

$$\varphi_{tot} = \oint \vec{D} \cdot \overrightarrow{dS} = D\,(4\pi r^2)$$

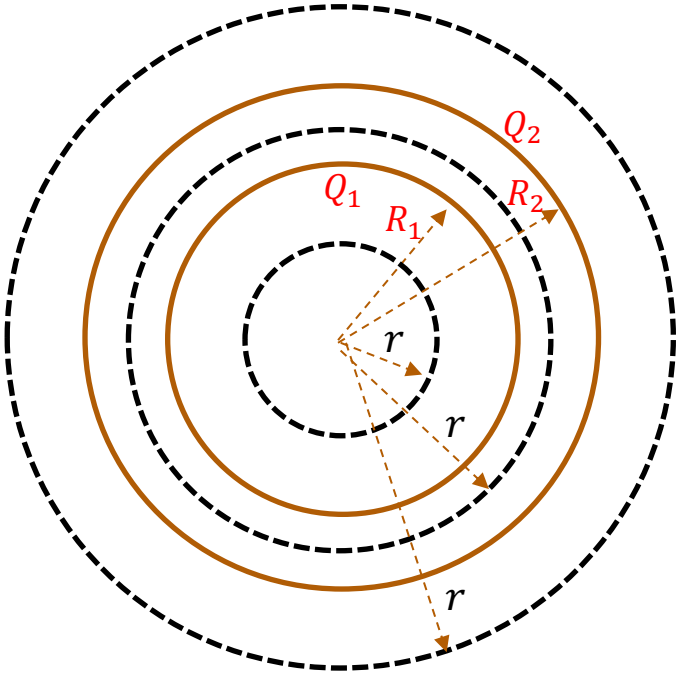
$$\rho_v \frac{4\pi}{3} R^3 = D\,(4\pi r^2) \qquad \vec{E} = \frac{\rho_v R^3}{3\varepsilon_o r^2}\,\vec{a}_r$$



Field Calculations using Gauss Law

Example. Two concentric conducting shells in free space with charges Q_1 and Q_2.

$$Q_{enclosed} = \varphi_{net} = \oint \vec{D} \cdot \overrightarrow{dS}$$



1) For $r \le R_1$

$$Q_{enclosed} = 0 \qquad \vec{E} = 0$$

2) For $R_1 \le r \le R_2$

$$Q_{enclosed} = Q_1 \qquad \vec{D} = \frac{Q_1}{4\pi r^2}\vec{a}_r$$

$$\vec{E} = \frac{Q_1}{4\pi\varepsilon_o r^2}\vec{a}_r$$

3) For $r \ge R_2$

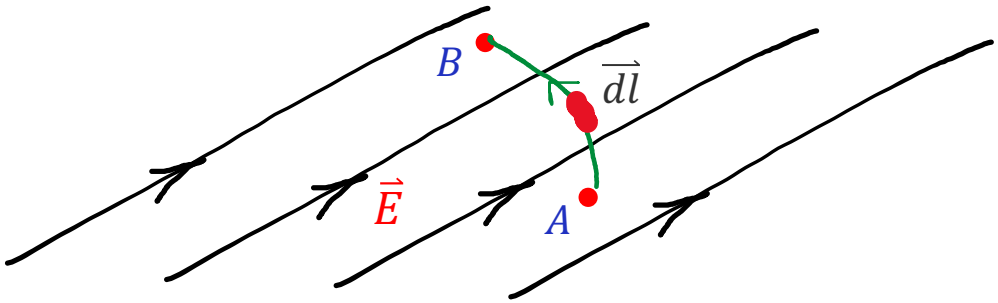
$$Q_{enclosed} = Q_1 + Q_2 \qquad \vec{D} = \frac{Q_1 + Q_2}{4\pi r^2}\vec{a}_r$$

$$\vec{E} = \frac{Q_1 + Q_2}{4\pi\varepsilon_o r^2}\vec{a}_r$$

The Electric Potential

The electric field can also be related and evaluated from the concept of potential energy.

Starting with the electrostatic force relation, we can find the work done on a charge moving within some electric field distribution.



$$\vec{F} = q\vec{E} \quad \rightarrow \quad dW = -\vec{F}\cdot\vec{dl} = -q\left(\vec{E}\cdot\vec{dl}\right)$$

The total work done by the field (energy gained or lost), going from point A to point B, is

$$W = -q\int_A^B \vec{E}\cdot\vec{dl}$$

From which, the electric potential is defined, as

$$V_{AB} = W/q = -\int_A^B \vec{E}\cdot\vec{dl}$$

Because the electric force is conservative,

$$V_{AB} = -V_{BA} \rightarrow V_{AB} + V_{BA} = 0$$

$$\oint \vec{E}\cdot\vec{dl} = 0 \quad \rightarrow \quad \vec{\nabla}\times\vec{E} = 0$$

Stokes Theorem

And, $\vec{E} = -\vec{\nabla}V$

The change in potential speaks about the properties of the field.

Q & A

Send me an email if you have any questions.